

CIND820: Initial Results

Project Name: Predicting Success of Video Games

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1 Data Analysis

1.1 Descriptive Statistics

The raw dataset used in this study contains historical information on video game titles with the variables and descriptions from Kaggle [1]:

- Rank - The rank of the video game based on global sales volume. (Numerical)
- Game Title - The name of the video game. (String)
- Platform - The platform on which the game is available, such as PC, PS4, xBox One, etc. (Categorical)
- Year - The year in which the game was released. (Date)
- Genre - The genre of the game, such as action, adventure, racing, etc. (Categorical)
- Publisher - The company that published the game. (String)
- North America - The number of units sold in North America, in millions. (Numerical)
- Europe - The number of units sold in Europe, in millions. (Numerical)
- Japan - The number of units sold in Japan, in millions. (Numerical)
- Rest of World - The number of units sold in the rest of the world, excluding North America, Europe, and Japan, in millions. (Numerical)
- Global - The total number of units sold worldwide, in millions. (Numerical)
- Review - The review score of the game, on a scale of 1 to 10. (Numerical)

Below is a summary of the key statistics from the dataset:

- 1907 data entries
- 22 unique platforms

- 12 unique genres
- 1.53 million median global sales
- 2.49 million average global sales
- 79.03 average review score
- 81.00 median review score
- Games' release date range from 1983 to 2012

1.2 Visualizations and Trends

It is important to note that sale values of 0 were excluded from each visualization under the assumption that the game was not released in that region.

Analysis of the dataset and visualizations shows a few key patterns. By looking at *Figure 1*, it can be seen that regional sales across North America, Europe, Japan, and the Rest of World are all strongly positively correlated, which is expected since global sales are the sum of these regions. *Figure 2* indicate that review score are not strongly correlated with global sales because it shows a high concentration of high-scoring games with relatively low sales in the bottom-right region which suggests that high review scores do not necessarily translate to high commercial performance. While *Figure 1* and *Figure 2* agree with each other, *Figure 3* seems to disagree with *Figure 1*. While the video game industry has grown over time with more units sold in recent years and with people buying more games as the years go on [2][3], the correlation with year and global sales is much weaker than expected.

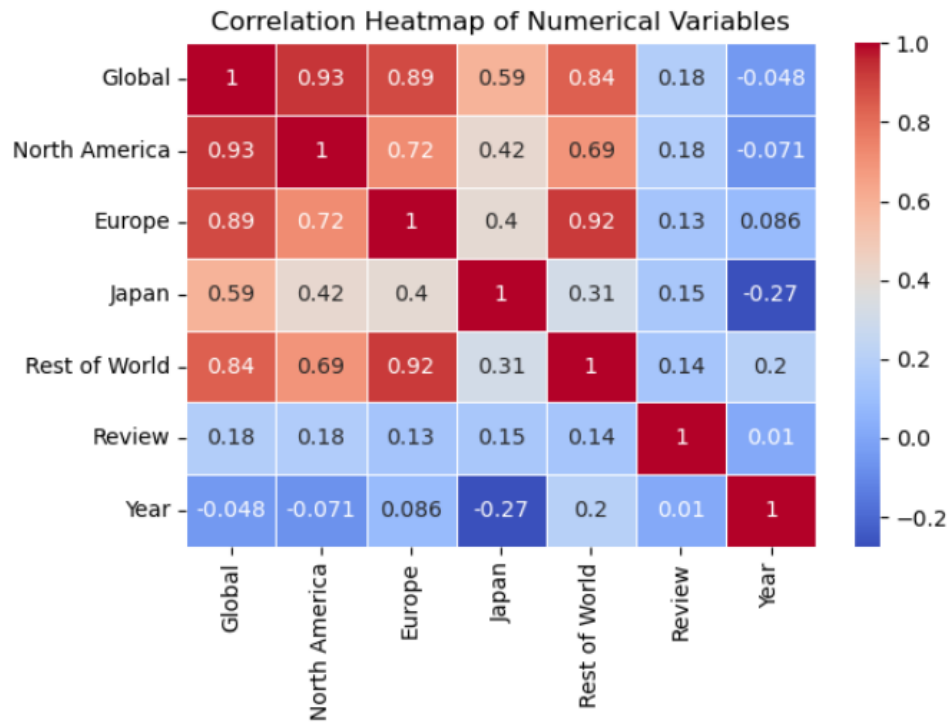


Figure 1: Correlation Heatmap

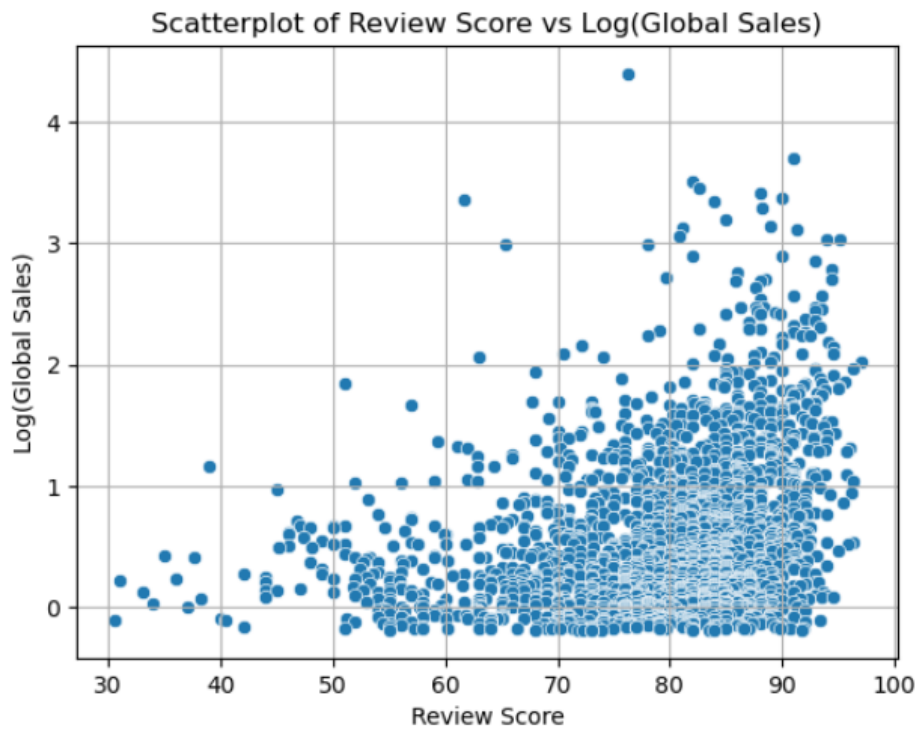


Figure 2: Log(Global Sales vs Review Score)



Figure 3: Global Sales vs Year

The genre and platform visualizations highlight a few notable outliers in the dataset. The median sales are skewed as seen in *Figure 6*, *Figure 10*, and *Figure 11*. Most games sold have similar sales performance when in all regions when grouped by genre and platform. Japan has drastically difference sales performance when it comes to games sold by platform and genre and the Rest of the World when it comes to platform. This indicates that developers/publishers that plan to market to a Japanese market or a Japanese developer/publisher marketing outside of Japan should do extra market research on their target audience [4]. When it comes to marketing to the rest of the world it likely depends on what console is popular in that region and what legislation causes specific games to become region locked [5].

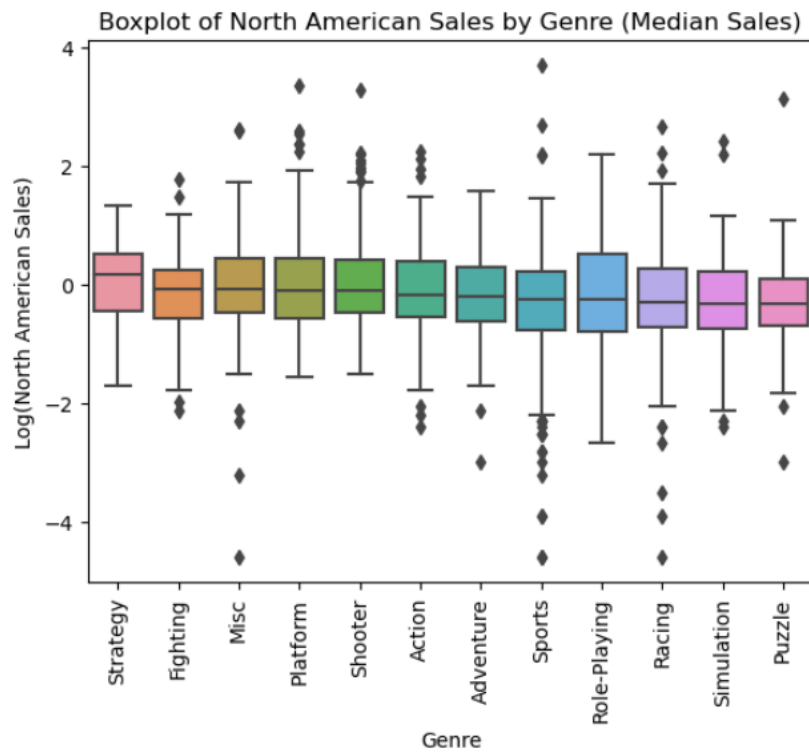


Figure 4: North American Sales Boxplot by Genre

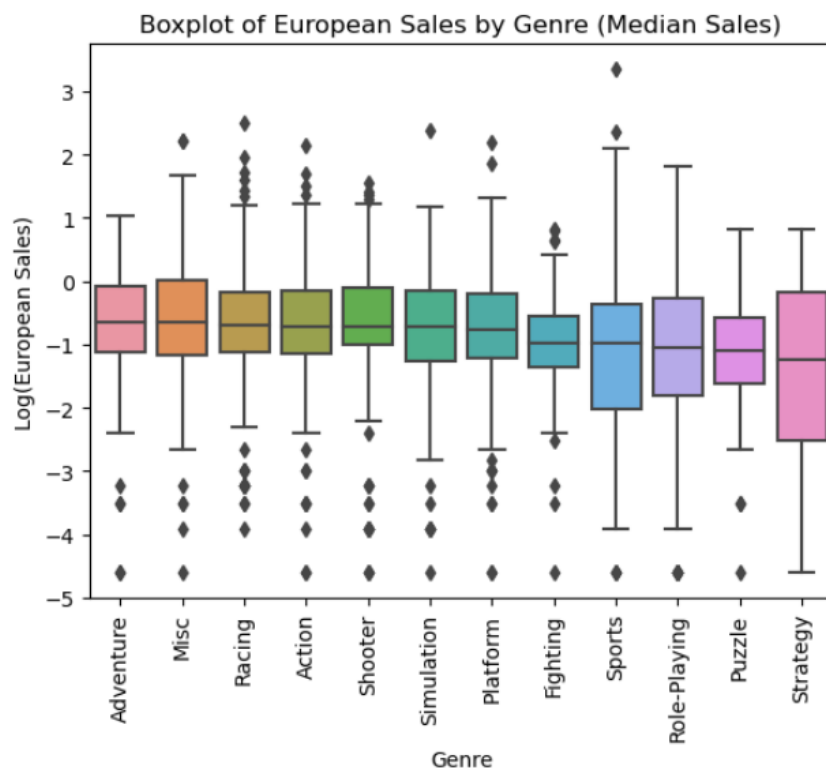


Figure 5: European Sales Boxplot by Genre

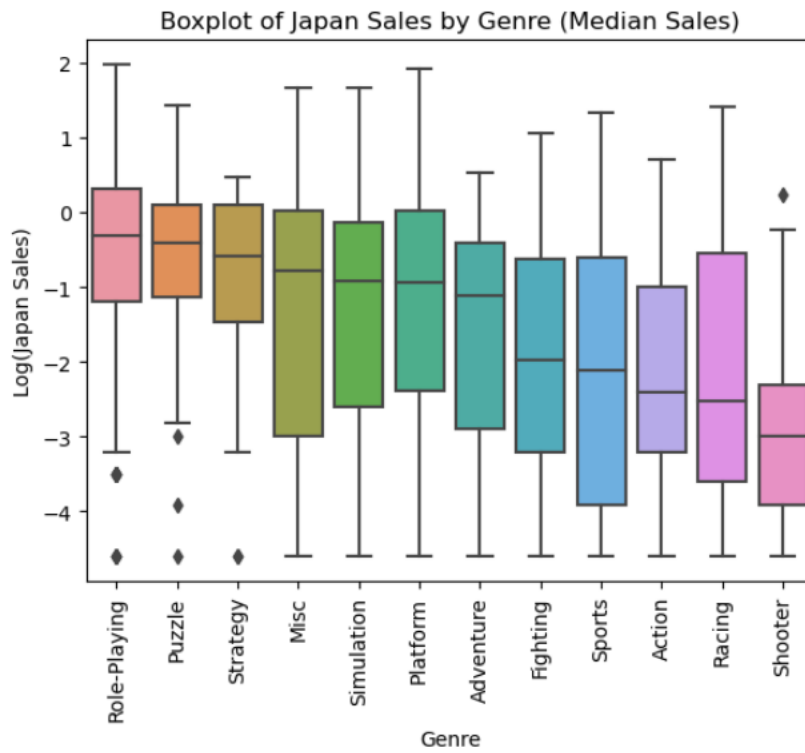


Figure 6: Japanese Sales Boxplot by Genre

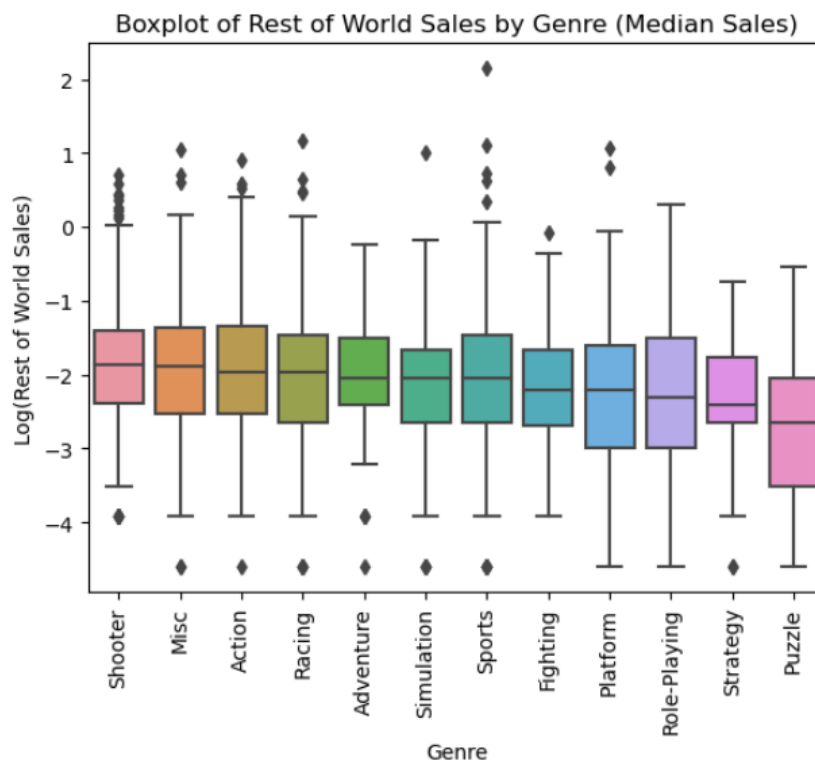


Figure 7: Rest of World Sales Boxplot by Genre

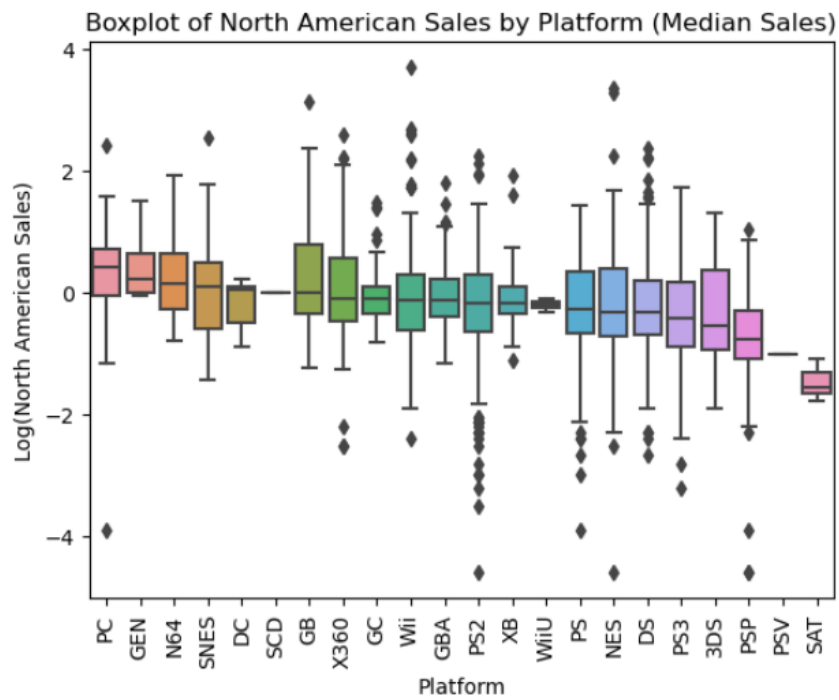


Figure 8: North American Sales Boxplot by Platform

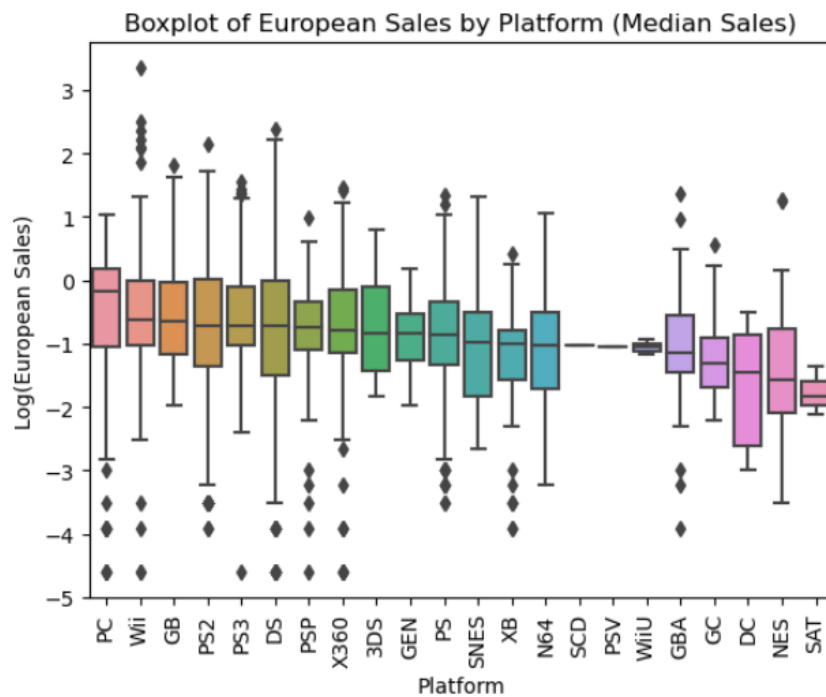


Figure 9: European Sales Boxplot by Platform

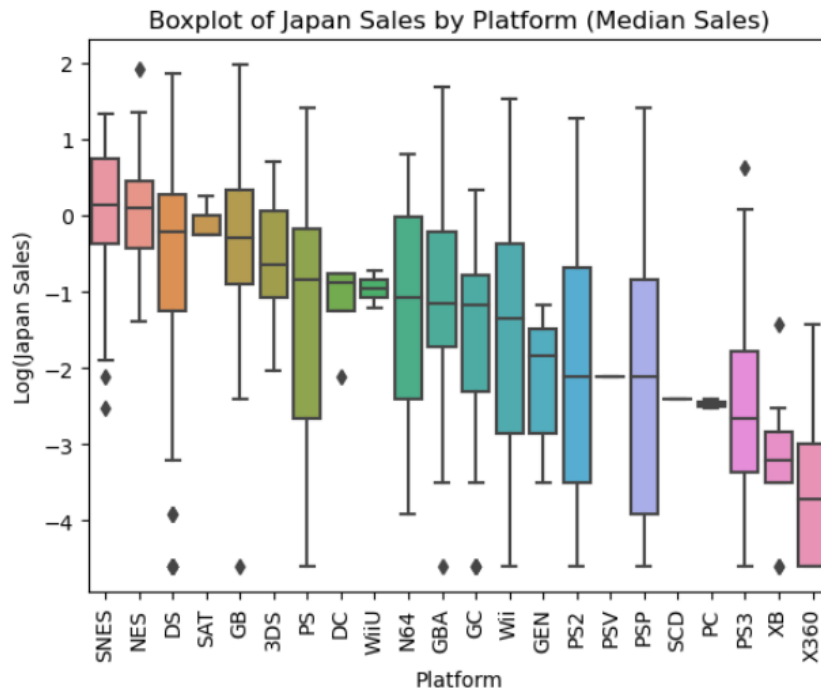


Figure 10: Japanese Sales Boxplot by Platform

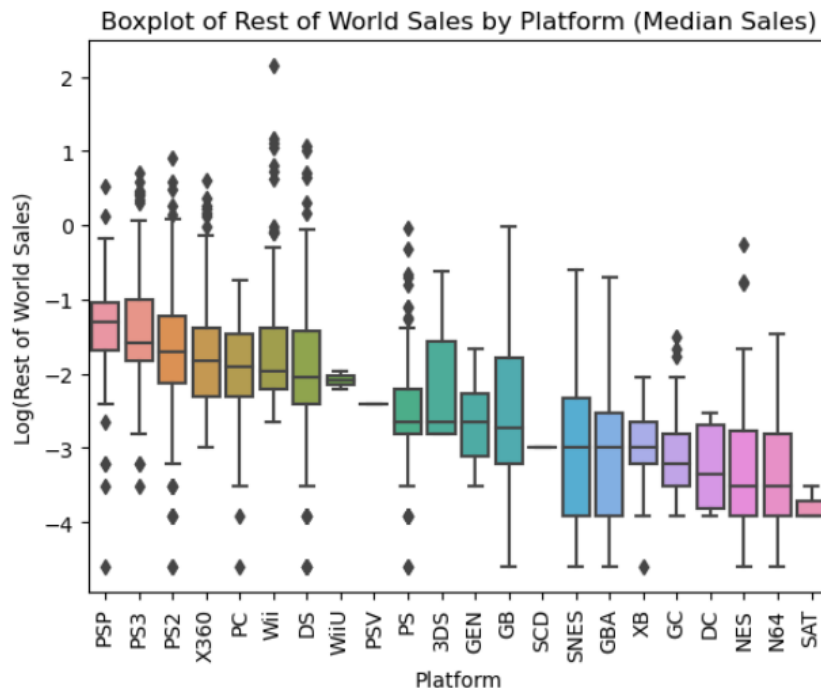


Figure 11: Rest of World Sales Boxplot by Platform

2 Data Preparation

2.1 Data Preprocessing

The data preprocessing steps and results are taken are as follows:

- Remove columns that contain a missing "Year" value
- Convert year from float64 to int64
- Convert object data types to string
- Round the float64 columns to 2 decimal places
- Categorical variables (Platform and Genre) were converted into numerical format using one-hot encoding
- Numerical features (Review Score and Year) were standardized using Standard-Scaler

2.2 Model Preparation

For predictive modeling, the preprocessed dataset was split into training (80%) and testing (20%) sets using `train_test_split` with a fixed `random_state` of 67 for reproducibility. The data was set up for a 5-fold cross-validation, which allows each model to be trained and evaluated on multiple splits of the dataset to provide a more accurate estimate of model performance and reduce overfitting. This provided a consistent input for the machine learning models used, which were: Random Forest, Gradient Boosting, K-Nearest Neighbor, Artificial Neural Network, and Support Vector Regression.

3 Model Evaluation

3.1 Model Development

The model development process consisted of feature selection, preprocessing, model training, and performance evaluation. The objective of the models was to predict global

video game sales using several explanatory variables from the dataset.

For feature selection, four variables were selected as input variables: platform, genre, review score, and release year. Platform and genre were categorical variables and were converted into numerical format with *one-hot encoding*, which creates binary variables representing each category [6].

Before training the models, the feature variables were standardized using *scaling* to ensure that variables the magnitude of each variable were not vastly disproportionate. Standardizing before modeling reduces influence on model performance which is particularly important for algorithms such as K-Nearest Neighbors and Support Vector Machines [7].

For training, the dataset was split into a training and testing set using an 80/20 split. The training set was used to train the models, while the testing set was used to evaluate the machine learning models. Splitting the data into training and testing sets prevents overfitting and ensures that models will apply to any new data given to the machine learning model [8].

To obtain a more reliable estimate for each machine learning model a 5-fold cross validation was applied. The dataset was divided into five random subsets and each subset was used once as a validation set while the remaining subsets were used for training. After each subset has a turn being evaluated by the machine learning models this way, the evaluation metrics were then averaged across the folds to produce a more stable estimate of model performance [9].

3.2 Model Evaluation

Five machine learning algorithms were evaluated: Random Forest Regression, Gradient Boosting Regression, K-Nearest Neighbors Regression, Artificial Neural Networks, and Support Vector Regression. Model performance was assessed using three evaluation metrics commonly used in regression analysis [6][7]:

- R^2 (Coefficient of Determination) - Measures the proportion of variance. Values closer to 1 indicate a better model and negative values indicate that the model

performs worse than simply predicting the mean of the target variable.

- RMSE (Root Mean Squared Error) - Measures the average magnitude of prediction errors and penalizes larger errors more heavily due to the squaring.
- MAE (Mean Absolute Error) - Measures the average absolute difference between predicted and actual values.

These metrics were calculated for a normal train-test split. They were also calculated for each fold during cross validation and then averaged to provide an overall estimate of model performance.

3.3 Initial Findings and Results

Table 1: Performance Metrics for Default Split of Nonlinear Machine Learning Models

Model	R ²	RMSE	MAE
Random Forest	0.1841	3.0729	1.5967
Gradient Boosting	0.1003	3.2268	1.6366
K-Nearest Neighbors (KNN)	0.1071	3.2145	1.6477
Artificial Neural Network (ANN)	0.2158	3.0126	1.6605
Support Vector Regression (SVR)	-0.0109	3.4204	1.3777

Table 2: Performance Metrics for Cross-Validated Non-Linear Machine Learning Models

Model	R ²	RMSE	MAE
Random Forest	-0.0459	3.4398	1.5423
Gradient Boosting	-0.0802	3.5100	1.5554
K-Nearest Neighbors (KNN)	-0.0310	3.4792	1.5875
Artificial Neural Network (ANN)	-0.2393	3.7540	1.7551
Support Vector Regression (SVR)	-0.0098	3.4747	1.3176

The evaluation metrics show important differences in how well each model predicts global video game sales. For the regular train-test split, the top performer is the artificial

neural network (ANN) with the highest R^2 of 0.2158, the lowest RMSE of 3.0126, and an average MAE of 1.6605. Random Forest performed similarly well with a slightly lower R^2 and RMSE but a similar MAE. Gradient boosting and K-nearest neighbors performed worse with much lower R^2 values even though their RMSE values were lower and had similar MAE values. The worst of these was the support vector regression model (SVR). It was the only model to end up with a negative R^2 value and it should not be used to estimate global video game sales at all. Overall, the regular train-test split results show that ensemble models like random forest and gradient boosting, as well as artificial neural network (ANN), are better at capturing patterns in this dataset than K-nearest neighbor or support vector regression (SVR) [7][6].

When the models were evaluated using 5-fold cross-validation, the performance decreased across the board. Each machine learning model ended up with a negative R^2 value compared to simply doing a regular train-test split, the only machine learning model with similar results was support vector regression (SVR). However, the cross-validated versions of the models are more similar to each other than the non-cross-validated versions aside from artificial neural network (ANN). The remaining 4 machine learning models all have very similar R^2 , RMSE, and MAE values. This shows that while some models perform well on a single subset, they may not generalize well. Overall, ensemble models with cross-validation seem to be accurate way to identify non-linear patterns.

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